Experience, Memory, and the Timing of Disaster Preparation*

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Abstract

Natural disasters have become more frequent and severe causing some people to experience them multiple times in quick succession. Additionally, changes in weather patterns and atmospheric conditions have made previously predictable aspects of natural disasters less so. While increasing experience with disasters be give people opportunities to learn how to prepare, any long lapses between subsequent disasters be result in discounting of the disaster's severity. I study the effects of experience and discounting on the average response to natural disasters by analyzing sales of emergency supplies, namely bottle water, during the 2008–2019 Atlantic hurricane seasons. Through event study analysis, I find that increases in historical exposure have little effect on the responsiveness of consumers; however, consumers respond increasingly to current hurricanes as the length of time since the most recent hurricane increases.

1 Introduction

Natural disasters are becoming more frequent and severe (Anderson and Bausch, 2006; Van Aalst, 2006). This trend has led to broad impacts on the economy as natural disasters cause poorer labor

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market outcomes, decreased land values, stalls in international trade, and increased government transfer payments (Ibarrarán et al., 2009; Deryugina, 2017; Deryugina et al., 2018; Sytsma, 2020). As disasters become more frequent, individuals living in threatened areas be experience an increasing number of disasters during their lifespan. This frequency allows people more opportunities to learn from disasters and use that information when preparing for future disasters. However, as storm patterns change, people who live in areas which previously experienced few storms be face disasters without having had prior learning opportunities. This lack of experience and information could lead to people not properly preparing when a disaster finally affects them. Additionally, because many years can pass before a person faces a subsequent disaster, the average person be face a discounting problem. If significant time has passed, the people be discount the effect of disasters and by extension, their need to remain prepared. The destruction caused by natural disasters can pose difficulties and safety hazards for people purchasing emergency supplies after the disaster. Hazardous road conditions after natural disasters can also make it more difficult for relief efforts to provide support for those who were not able to properly prepare (Horner and Widener, 2011). In some areas, a lack of experience with storms or extreme discounting due to a lack of recent disasters be lead to an increase in the required funds and efforts needed to restore the area. Thus, it is important to understand how experience and discounting of disasters influences preparation in the future. By identifying any gaps in responsiveness and the mechanisms that drive them, firms can better prepare for the response of their customers and avoid any shortages or supply chain issues that be arise. Understanding the mechanisms driving the response to natural disasters would also allow government agencies to make and propose policies that target the issues before hand, rather than waiting to respond after the fact.

In this paper, I explore how experience with hurricanes and the time between subsequent hurricanes affects purchasing patterns of emergency supplies in the Atlantic and Gulf Coast regions of the United States (i.e. areas most often subject to the annual Atlantic hurricane season). To understand this relationship, I combine NielsenIQ Scanner Data with geographic forecast data from the National Oceanic and Atmospheric Association (NOAA) from 2008–2019. With modern forecasting techniques, consumers can respond to the threat of a hurricane days before it lands.[1] This allows them to purchase emergency supplies in response to the forecast rather than the hurricane itself. Using the scanner data I observe the sale of emergency supplies throughout the year for nearly all counties in the Southeast. Additionally, I employ NOAA's historical hurricane landfall data (HURDAT2) in order to build out each county's hurricane history.

My analysis begins with a baseline exploration of the effect of hurricane forecasts and landfalls on the sales of different emergency supplies in the style of Beatty et al. (2019), before deviating to explore the effects of historical, county specific attributes. I focus solely on sales of purified drinking water. While there are many items on the Department of Homeland Security's emergency kit checklist, I believe water most easily captures the responsiveness of the population. The reasons for this choice are that water is the first item on the checklist, it is the most homogeneous product on the list, it is readily available at nearly every store nation-wide, and it is often in the news and social media during natural disasters. I find that not only do sales of bottled water increase from 20–30% during the weeks surrounding any general hurricane, but I also find that this result is entirely driven by storms classified as a category 1 hurricane or higher.

I then diverge from their analysis by exploring the effects of county specific historical experience variables on sales of emergency supplies during a current hurricane. The first historical variable I analyze is the county's total historical landfall count since 1983. This allows me to test for the possibility of learning behavior. I analyze the responce to a current hurricane in counties with different levels of past hurricane landfalls. While migration and other factorrs be change the makeup of the county. The average person living in a high exposure area will likely have a different understanding of the general risk of hurricanes compared to people in less esposed regions. While the event studies do not show any significant quartile differences, the regression results show that additional hurricane exposure leads to roughly an additional 6% inrease in sales durign a hurricane forecast.

I next analyze the effects of the length of time that passes between subsequent hurricanes. If a

[1] Hurricane seasons produce cyclonic storms of different intensities, including storms labeled by NOAA as "disturbances", "tropical depressions", "tropical storms", and "hurricanes", all of which can cause damage to infrastructure and housing. I primarily focus on the effects of hurricanes and major hurricanes and thus only use the term "hurricanes".

county has not been hit for many years, regardless of its historical exposure, people's perceptions of the risk be start to fade. Through event study analysis I find a weak pattern indicating that the sales of bottled water during a hurricane increase as the time since the last hurricane increases up to 6 years. However, after the 7th year, the sales begin to decrease. The regression analyzes the impact of one additional year between subsequent landfalls. While the linear model show a significant increase in sales close to a hurricane landfall of roughly 13%, the results of the quadratic model are insignificant.

Much of the current literature on natural disasters focuses on changes in markets after a disaster. Such literature includes effects of insurance take-up rates, labor market impacts, and government assistance applications (Hallstrom and Smith, 2005; Gallagher, 2014; McCoy and Walsh, 2018; Bakkensen et al., 2019; Groen and Polivka, 2008; Deryugina, 2017; Deryugina et al., 2018). Others focus on how markets and decisions are made in the days leading up to a singular disaster through the use of phone surveys (Kelly et al., 2012; Meyer et al., 2014). I focus on the intersection of this literature by studying how past experiences with hurricanes affect the responses in the current period. I find that the historical exposure leads to an increase in early purchases of emergency supplies.

The literature on the salience of disasters tends to use variables with lower purchasing frequencies such as houses and insurance or changes in financial markets to understand how recent disasters affect decisions (McCoy and Walsh, 2018; Bakkensen et al., 2019; Bourdeau-Brien and Kryzanowski, 2020). Others that focus on a more individual response rely on either general surveys or surveys after disasters to understand feelings around risk and preparation steps that had been taken (Lazo et al., 2010; Baker, 2011; Meyer et al., 2014). The closest research to mine is that in the wildfire literature which looks at people's interest in home air purifiers, and the decision to stay indoors (Burke et al., 2022). I add to this literature by utilizing sales data of emergency supplies that are more liquid than a home or insurance and can be purchased much more frequently. Additionally, emergency supplies are relatively inexpensive, allowing me to capture the salience of hurricanes for a larger portion of the affected population. My findings follow the pattern that behavioral responses tend to fall off after six years. I also find that this leads to an increase in purchases closer to the time of landfall.

The rest of the paper proceeds as follows. Section II discusses the background of hurricane preparation. Section III explains the data followed by the methodology is section IV. Main results are found in section V while section VI concludes.

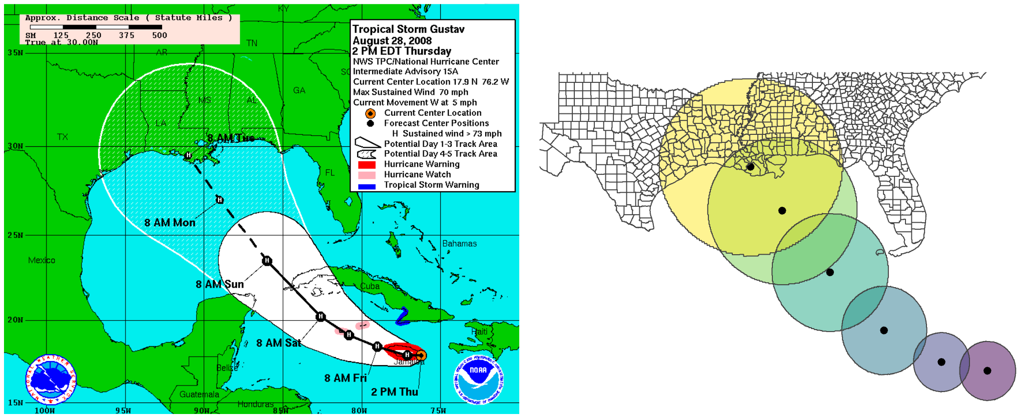
2 Background

The Department of Homeland Security (DHS) along with other federal, state, and local government agencies distribute information on how to prepare for a myriad of natural disasters. Part of preparing for such events is the building a "Disaster Supply Kit". The list of emergency supplies to include in each kit is identical among agencies, as they all use the DHS checklist as a guide. The specifics of this list can be found at the DHS website, ready.gov. This checklist includes items such as purified drinking water, non-perishable foods, batteries, flashlights, first-aid kits, and many more.

When signs of a tropical system (potential hurricane) first appear, the National Oceanic and Atmospheric Administration (NOAA) begins tracking the system. NOAA gathers information on the size, wind speed, movement speed, ocean temperature, and other atmospheric conditions that will affect the tropical system. Using this information, they begin forecasting the system for the next 5 days. Specifically they forecast the expected wind speed and location along with storm surge, tropical storm watches and warnings, and hurricane watches and warnings (NOAA, 2024). To keep the populous well-informed on the formation and expected path of the system, NOAA posts a new forecast advisory every six hours.

Because of the frequent updates, as well as modern technology providing quick access to information, people have time to respond to the threat of an impending hurricane well before it ever makes landfall. This provides the unique opportunity to observe how the population responds to such information. Additionally, we can observe how prior experiences and knowledge play a role in how people utilize the information they are being given in relation to disasters.

Figure 1: Replication of Official NOAA Forecast



Note: Maps of official and replicated forecasts of Hurricane Gustav (2008). Official advisory (left) is from August 28, 2008, at 2pm with the replication (right) made from the archived shape file of the same advisory. Gustav made initial landfall in Louisiana at 10pm on Monday, September 1, 2008.

3 Data

3.1 Hurricane Data

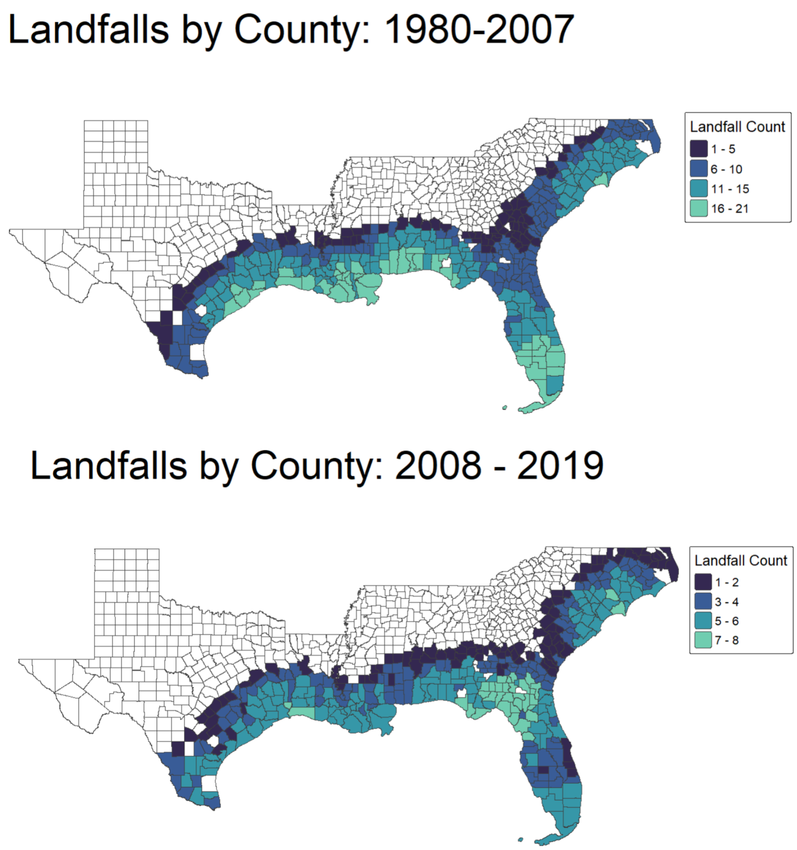
The National Hurricane Center (NHC) GIS Archive of Tropical Cyclone Forecasts contains all shape files used to build the hurricane forecasts and watch/warning advisories since the 2008 Atlantic Hurricane Season. There is a new file for each advisory update that occurred: every six hours from the system's conception to its dissipation. Additionally, each file contains information on the system's current location and wind speed as well as the forecast paths, wind speed, and weather advisories for the next five days. Using this information, I am able to replicate the forecast as it appeared in the past. Figure 1 shows the results of using the archived shape files to replicate the forecast as it was seen by individuals in the past.

To build out the advisories I follow an approach similar to that of Beatty et al. (2019). By drawing a circle with a 100 nautical mile radius around the current location of the hurricane, I am able to identify any counties that are in the average affected area of any given hurricane. I then draw circles around the forecast locations in varying sizes. The circle around the one-day out location has a radius of 100 nautical miles, two days out has a radius of 150 nautical miles, three days out 200, 4 days out 250, and 5 days out 300. Through this I am able to replicate the forecast track and accompanying cone of uncertainty shown to the public during the initial forecast. This allows me to identify which counties are within the forecast, and at which point they be be affected by the impending hurricane. Because of the speed at which some hurricanes move, occasionally a county is inside of two forecast circles at the same time. While this be seem like a challenge at first glance, it does not pose any issues. Because the movement speed of hurricanes can speed up and slow down with the weather, it is possible for the forecast timing of when a hurricane is predicted to make landfall to shift. Additionally, because the advisories are updated every six hours, there are 4 observations for every day. This would result in counties often being in two different forecast circles even without overlap.

In addition to the GIS Archive, I also use NOAA's HURDAT2 (a.k.a. "Best Track") data. This data set contains the location and wind speed of all recorded hurricanes since the early 1900s in six hour increments. Because hurricane tracking did not become consistent until more recently, I only use the data since the 1980s as recommended by NOAA. By combining these two data sets, I observe a complete history of hurricanes since the 1980s. Figure 2 shows the total count of hurricanes to make landfall in each county from 1980 to 2007, during which I only have the HURDAT2 data, and from 2008 to 2019, once the GIS Archive is made available. It can be observed from these maps which specific regions of the Southeast are hot-spots for hurricane landfalls. Interestingly, historical landfall counts do not perfectly correlate with the most recent decade of landfall. Specifically with the Gulf Coast from Texas to Alabama seeing fewer hurricanes, while North Florida and Georgia became the most affected by hurricanes.

From combining the HURDAT2 and GIS Archive, I create two new variables to capture different aspects of a county's history with hurricanes. The first variable I create is the total historical landfall count of each county, $HurCount$. This variable will capture the impact of living in an area that is more geographically prone to hurricane landfalls. Additionally, this variable will capture any effects of permanent shifts in hurricane landfall locations due to climate change. To build this variable, I first use the HURDAT2 data to set a baseline for each county. The total number of hurricanes to make landfall in each county from 1980 to 2007 is used as their starting hurricane count on January

Figure 2: Total Hurricane Landfalls

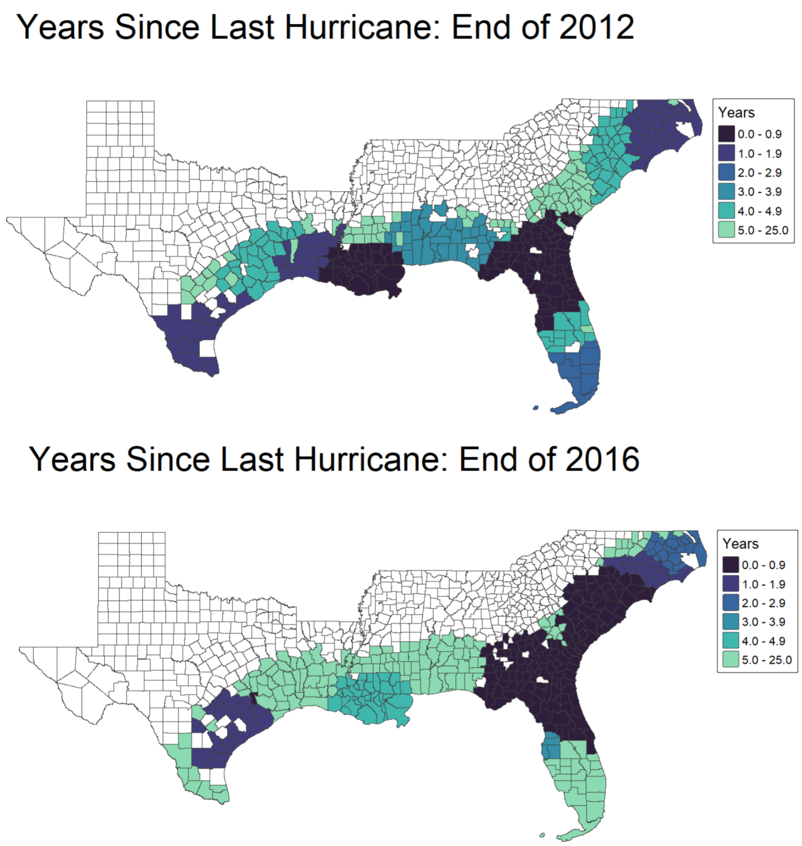


Note: Maps of official total tropical systems to make landfall from 1980 to 2007 (top) and from 2008 to 2019 (bottom). The lighter colors represent more landfalls during the given time frame.

1, 2008. The relative starting values are observed in the top panel of figure 2. At the beginning of each following year, the sum of landfalls in the year prior are then added to the total landfall count.

The second variable I create from this combined data is the length of time that passes between subsequent landfalls, $YrSince$. This variable captures the effects of the cyclical nature of hurricanes season from the El Niño and La Niña cycles. Due to this cycle hurricane seasons go through subsequent years of high hurricane counts followed by years of low hurricane counts. This can lead to areas which are more geographically prone to hurricanes to go years without any hurricane

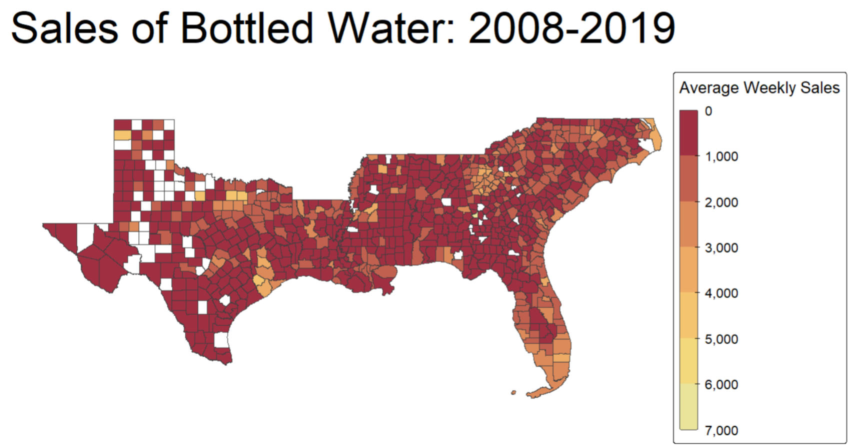
Figure 3: Variation in Timing of Hurricane Landfalls



Note: Maps representing the years that have passed since that last hurricane made landfall in a county. Each map is based on the hurricane record during the last week of the year. Thus, counties with 0 years since landfall in the top map were hit by a hurricane sometime in 2012.

activity or vice versa. To build this variable I use the HURDAT2 data to find the hurricane that hit each county most recently to January 1, 2008. Each entry in the data adds another week since the prior hurricane landfall. A week is added to this variable for each entry until the county sees a new landfall. After a county is hit by a new hurricane, the time since the prior storm is reset to account for the new landfall. Figure 3 shows this timing variable at the end of both 2012 and 2016.

Figure 4: Average Sales of Bottled Water



Note: A map of the average total revenue from bottled water sales per store in each county from 2008 to 2019.

3.2 Scanner Data

Using NielsenIQ Retail Scanner Data, I am able to observe weekly level sales of emergency supplies and other goods across the Southeast. This data contains weekly sales of all goods sold at each reporting store. In addition to the quantity sold, the data contains information on the price, any sales or specials, the sizes of products, and the county that the store is located in. Although the data set covers the entire nation, I choose to limit my analysis to those states most often affected by the Atlantic hurricane season[2]. Additionally, I limit the analysis to only sales of bottled water. I choose bottled water over other products first because it is the first item listed on the official emergency kit checklist. Second, because its low price makes it affordable to the vast majority of consumers. Third, it is a rather homogeneous good that relies less on tastes and preferences than more broad category items like non-perishable foods. Finally, because the scanner data primarily covers grocery type stores, giving much more coverage to water than goods such as flashlights.

In order to understand how the population as a whole is responding to hurricane forecast, I aggregate the data to a county-week level. First I aggregate the sales of individual bottled water products in each store by calculating total revenue from bottled water sales and average price

[2]States included are Alabama, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, and Texas.

of all water sold. Then I combine the store level sales to the county by calculating the average total revenue per store and the average price per store. I must use the average revenue per store in a county rather than the total revenue per county because only select stores are represented in the data. Without this weighting, some counties would be over or under-represented in their responsiveness to hurricane forecasts. The average weekly total revenue from bottled water sales for each county can be seen in figure 4. This figure shows that Atlanta, Dallas-Forth Worth, Houston, and the Atlantic Coast areas purchase the most bottled water on average.

3.3 Control Data

Because bottled water sales can be affected by high heat days, I use data from NOAA's Global Historical Climatology Network. Average temperature is recorded at the daily level at hundreds of weather stations across each state. I first match each county to its closest weather stations, then I take the average of the temperature across the entire week. While this gets rid of some of the variation, it still provides enough to understand if the week was hotter than average.

There is also significant variation in the population of counties in the data. Due to this, average sales per store are likely to be significantly larger in more populous counties skewing any results. To correct for this, I use population data from the U.S. Census. Using the county level population from the 2010 census, I create a sales per 100,000 residents variable.

4 Methodology

There are two dominant mechanisms for historical experience to play a role in the level of "panic buying" that occurs during a hurricane forecast. The first is the accumulated knowledge of the current county population through both the knowledge of prior generations that is passed on, and the knowledge gained through personal experiences. Living in an area that is prone to a certain type of natural disaster allows people to learn about that specific disaster either through personal experiences or through the shared knowledge of those around them, regardless of how long they have lived in the area. As areas are affected by more and more hurricanes, the average knowledge of the population in those areas be increase following traditional learning models. This would likely reflect in sales of emergency supplies with the total sales during a hurricane forecast being at different "equilibrium" reaction levels based on the accumulated experiences.

The second mechanism likely to influence "panic buying" is the time that has passed since the last hurricane hit an area. Regardless of the baseline knowledge of a county, when a hurricane has just made landfall, the memory of the results is fresh. As time passes memory tends to fade. This memory likely also affects the responsiveness of consumers to hurricane forecasts. This affect may also be impacted by the strength of the last hurricane as well. If a recent storm was a tropical depression, it would likely not change the area's responsiveness since the damage and impact would be minimal. However, if the last hurricane were a major hurricane, the damage would be extensive, and its impact of the general population much larger. Conditional on the last hurricane strength, an increases in sales as time passes likely indicates that memory has a large effect on "panic buying".

While it may seem that these two variables, overall experience and years between subsequent hurricanes, are collinear in nature and cannot be separately identified, they are actually driven by separate forces. The total experience and therefore the likeliness of a county being hit by a hurricane during any given hurricane season is purely geographic in nature. South Florida around Miami juts out into the Atlantic Ocean setting it right in the path of hurricanes. However, North Florida around Jacksonville, is hardly ever hit due to the currents in the ocean and atmosphere pushing hurricanes farther north and east. The relative timing between subsequent events, or the intensity of subsequent hurricane seasons, is far more driven by meteorological conditions, namely the El Niño and La Niña cycle and the Atlantic Multi-Decadal Oscillation (NOAA, 2014). Specifically, being in a La Niña cycle makes weather conditions more favorable for hurricanes to form and remain active, thus increasing the likelihood of all coastal counties to be hit in the year. Vice versa, the El Niño cycle makes conditions unfavorable for hurricanes to form and stay active, thus reducing the chance of any coastal counties getting hit. One clear example of the difference in these two variable is seen in Miami-Dade County, Florida. Miami-Dade had 15 landfalls from 1980 to 2007, a relatively high number. Miami-Dade was hit by hurricanes in 2008 and 2010, a two-year gap, but was not hit again until 2017, a 7-year gap. It was then hit again the same year and the following.

4.1 Event Study Models

I begin my analysis by conducting several event studies to test for more general impacts of county histories with hurricanes. The event studies are conducted using the following regression:

$$Sales_{c,t} = \sum_{\tau=-10,\tau\neq-2}^{10} \beta_{\tau=t} L_{c,\tau=0} + \alpha_c + \alpha_m + \alpha_y + \gamma X'_{c,t} + \varepsilon_{c,t} \tag{1}$$

$Sales_{ct}$ represents the population-store weighted average total revenue from bottled water sales in a given county-week. The variable $L_{c,\tau=0}$ indicates if a tropical system or hurricane makes landfall in a county during the event week $\tau = 0$. I use week $\tau = -2$ as the reference period for the event study to avoid spillover from the forecast in week $\tau = -1$. Because the scanner data is reported at a weekly level, if a storm were to make landfall early in week $\tau = 0$ (i.e. Sunday or Monday), many of the purchases in advance would be done in the sales week before the storm lands, $\tau = -1$. The parameters of interest are $\beta_{\tau=-1}$, $\beta_{\tau=0}$, and $\beta_{\tau=1}$. Positive coefficients on $\tau = -1$ or 0 would indicate that individuals are waiting for a storm to purchase emergency supplies. A null result on these weeks would indicate that there is little evidence to support the "panic buying" is a consistent behavioral response. Average weekly temperature is controlled for in X'_{ct}. County, month, and year fixed effects are represented as α_c, α_m, and α_y respectively. I allow for clustering of the standard errors at the county level.

To test for any effect from changes in total experience or from changes in timing between subsequent storms, I subset the data into quartiles along the $HurCount$ variable and again along the $YrSince$ variable. The quartiles for $HurCount$ are 1 to 7, 8 to 12, 13 to 16, and 17 or more. Because the variable is updated at the beginning of each year, it is possible for counties to move between quartiles or even pass counties that started in higher quartiles. Quartiles for $YrSince$ vary depending on how the variable is defined. When $YrSince$ is the number of years between the last storm of any kind and the next hurricane the quartiles are less than 1 year, 1 to 2 years, 3 years, and 4 or more years. When $YrSince$ looks exclusively at the years that pass between subsequent hurricanes, the quartiles are 1 to 2 years, 3 years, 4 to 6 years, and 7 or more years. Again, because the variable is continuously updating, counties may fall into multiple quartiles throughout the panel.

4.2 Interacted Regression Models

After testing for quartile results with event studies, I move to regression analysis to test for incremental effects of experience with and timing between hurricanes. The model used in this portion of the analysis follows.

$$\begin{aligned} lnSales_{c,t} =& \beta_1 Threat_{c,t} + \beta_2 Landfall_{c,t} + \\ & \beta_3 H + \beta_4 H \times Threat_{c,t} + \beta_5 H \times Landfall_{c,t} + \\ & \alpha_c + \alpha_m + \alpha_y + \gamma X'_{c,t} + \varepsilon_{c,t} \end{aligned} \tag{2}$$

The log of population-store weighted average total revenue from bottled water sales for a given county-week is represented by $lnSales_{c,t}$. $Threat$ indicates if a county was within the "cone of uncertainty" during any of the forecasts for that week. $Landfall$ indicates if a county was within the average landfall radius of a hurricane eye during the given week. The variable H represents the historic hurricane related variables of interest, $HurCount$ and $YrSince$. Weekly temperature is controlled for in X'. Additionally, there are county level fixed effects, α_c, as well as both year, α_y, and month, α_m to capture any seasonality in sales. The variables of interest are β_4 and β_5 as these will indicate the effects of historical hurricane trends on "panic buying" related sales specifically. Standard errors are clustered by county-year.

4.3 Robustness

Significant positive results of the coefficients from the linear model of years between are not inherently causal. Purchasing patterns could increase for a myriad of reasons beyond fading memory. Sales may also increase as years between subsequent hurricanes increases because people's purchasing changes to early in the hurricane season and then slowly changes back to a "panic" response. Sales could also increase because people have a stockpile from the prior hurricane that runs out over time, and the more time that passes, the more likely they are to need to restock.

The first way to test for causality is to test if the change in response follows that of prior literature. Past literature on natural disasters and memory effects find a consistent result of an initial reaction followed by a dropoff in responsiveness (Gallagher, 2014; Kelly et al., 2012; Meyer et al., 2014). I test for changes in responsiveness in two ways. First, the event study model provides a visual test for any patterns in responsiveness. Second, I add a quadratic interaction in equation 2 so that the new model is as follows.

$$
\begin{aligned}
lnSales_{c,t} = &\beta_1 Threat_{c,t} + \beta_2 Landfall_{c,t} + \beta_3 H + \\
&\beta_4 H \times Threat_{c,t} + \beta_5 H \times Landfall_{c,t} + \\
&\beta_6 H^2 \times Threat_{c,t} + \beta_7 H^2 \times Landfall_{c,t} + \\
&\alpha_c + \alpha_m + \alpha_y + \gamma X'_{c,t} + \varepsilon_{c,t}
\end{aligned} \tag{3}
$$

If the quadratic terms are significant, then it is more likely that this is a behavioral response rather than one of the other mechanisms driving the change.

To further test for a causal relationship, I run the same event studies and regressions for the start of hurricane season (week containing June 1). Changes in sales during the start of the season would provide strong causal evidence that people are changing their behavior in response to the recentness of a hurricane. A lack of significant result at the start of the season indicate that it is likely some other cause.

Testing for current inventory holdings changes is beyond the scope of this paper. Because sales are only reported at a store-week level, it would not be possible to determine how much individual household level inventory changes. I analyze this effect using a different set of data in further research (Wood, 2026).

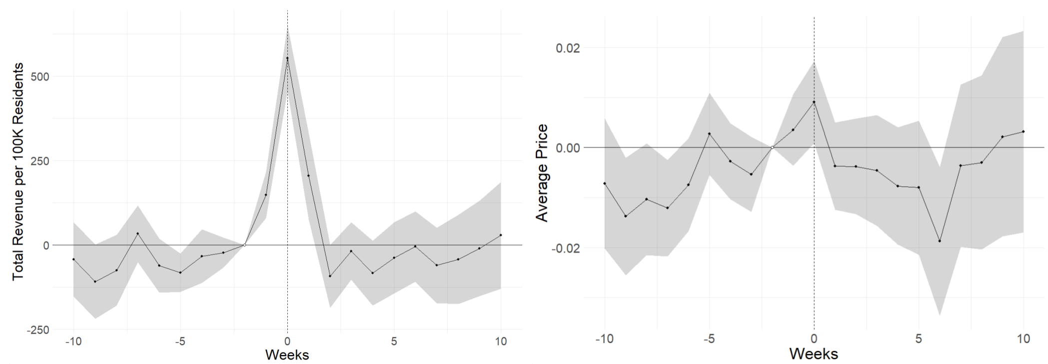
5 Results

5.1 Event Study Results

I begin my analysis by testing for aggregate effects of hurricane forecasts and landfalls. By running the event study from equation (1) on all counties, I estimate the average response of all counties to a hurricane. Figure 5 Panel (a) shows a clear sudden increase in sales of bottled water during

Figure 5: Aggregate Event Studies of Bottled Water Sales

(a) Bottled Water Sales (b) Price



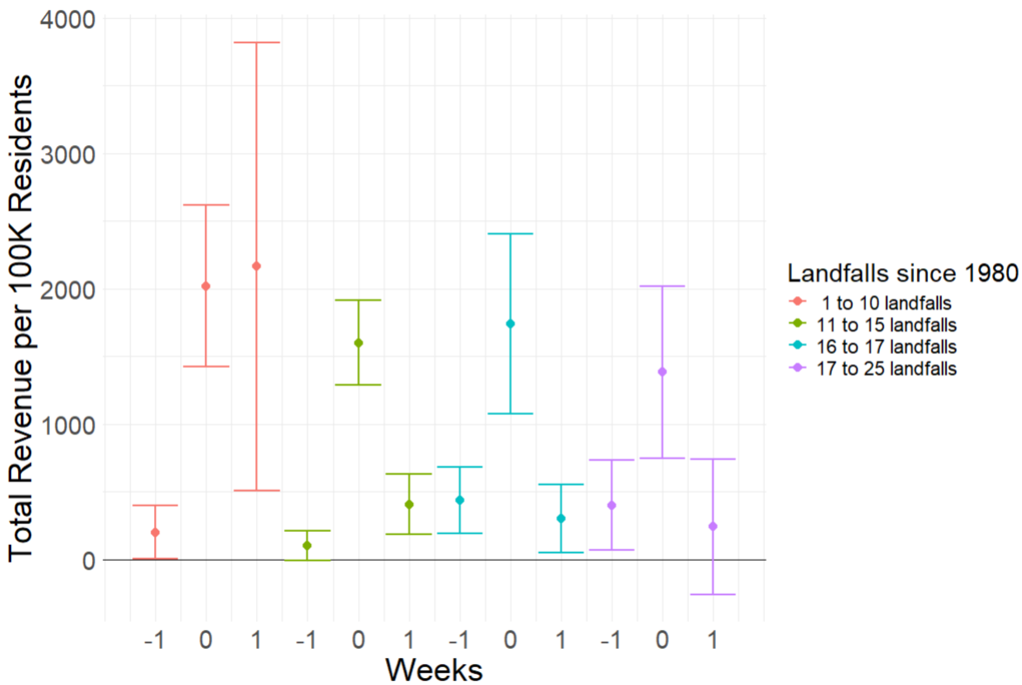
Note: Results from event studies following equation 1. The vertical axis represents total revenue per 100 thousand residents in USD in panel (a) and average weekly price in panel (b). The horizontal axis represents weeks. The reference period is week -2. The vertical dashed line at week 0 represents the week when a storm made landfall. The band around the estimates represents the 95% confidence interval.

the weeks surrounding a hurricane landfall with sales increasing by over \$500. Sales more than two weeks before and after the hurricane are not significantly different from zero indicating the effect of the hurricane is temporary in nature and not permanent. These results follow closely to those of prior literature, namely Beatty et al. (2019).

I also run the event study on the entire panel using price as the outcome variable rather than sales. Since the data is at a weekly level, price is the average price of bottled water in each county for the week. For any changes in sales responses to be behaviorally driven, it must come from a change in the amount that people are purchasing, not from changes in the pricing structure. Figure 5 panel (b) shows the results from the pricing event study. Here there are no significant changes in the price of bottled water, and the largest deviation seen during the hurricane is less than 1 cent. Thus, any changes in sales are likely from changes in consumer behavior.

The HURDAT2 and GIS data include all classifications of tropical systems, yet news and social media sources tend to focus on systems classified as hurricanes and major hurricanes as these are the most destructive. I rerun the event study subdividing the data by three broad hurricane classification categories: less than hurricane, hurricane category 1–5, and major hurricanes (category 3–5). The results of the event study by category are in figure **??**. It is clear from this figure that

Figure 6: Event Studies of Historical Exposure on Sales

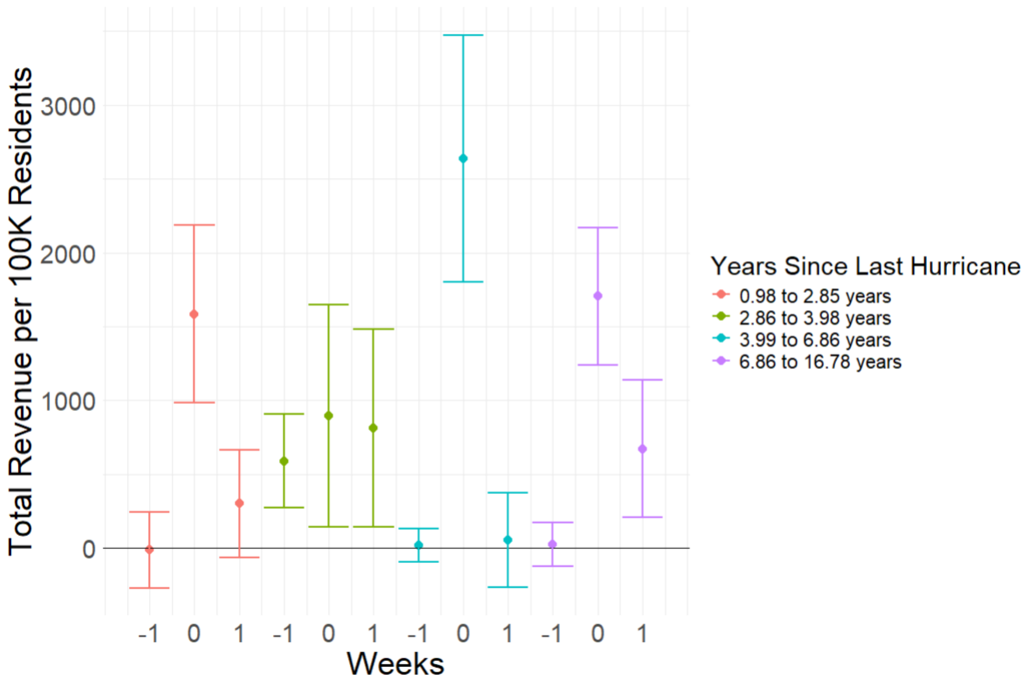


Note: Results from event study with subsets based on historical exposure of the county to hurricanes. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars represent the 95% confidence interval around the point estimate.

storms classified as a category 1 hurricane or stronger are driving all the results in Panel (a) of figure 5. Confidence intervals can be seen in figure A4 in the appendix and show that the effects of hurricane and major hurricane landfalls are significantly different from the results from a landfall of a lesser system. Additionally, the effect of a lesser system are not significantly different from no storm at all. For the remainder of the analysis, I only look at the effects hurricane specific landfall.

I test for general effects of living in a more hurricane prone location by subsetting the data along the variable $HurCount$. The data is subset into the following quartiles: 1 to 7 prior landfalls, 8 to 12, 13 to 16, and 17 or more. Figure 6 shows the results of the event study along quartiles. The weeks shown in the graph are those immediately surrounding the hurricane landfall. Results of each full event study are show in appendix figure A5. Consistent with the aggregate results, all quartiles

Figure 7: Event Studies of Timing on Sales



Note: Results from event study with subsets based on the years that have passed since that last storm made landfall in a county. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars represent the 95% confidence interval around the point estimate.

respond significantly to a hurricane landfall, but the responses from one quartile to the next are not significantly different from one another. Additionally, the estimates on the week before landfall are not significantly different from one another. From this analysis alone, there is not enough evidence to support the argument that hurricane exposure directly increases knowledge.

I test for aggregate effects of repeated high intensity hurricane seasons by repeating the event study from equation (1) with the panel subset by variable $YrSince$. The data is split into the following quartiles: 1 to 2 years between hurricanes, 3 year, 4 to 6 years, and 7 or more years. Figure 7 shows the results from this split of the data. The results seem to indicate a weak pattern of increased purchases as the number of years between subsequent years grows, eventually declining some time after year 6. The standard errors, however, are quite large resulting in estimates that

Table 1: Historical Exposure Results

Dependent Variable:		Log Sales	
Model:	(1)	(2)	(3)
Variables			
Threat	0.1073	0.1220	-0.8372**
	(0.1564)	(0.1670)	(0.3462)
Landfall	0.4481***	0.4157***	1.363***
	(0.0683)	(0.0650)	(0.1631)
Landfalls Since 1980		-0.0831***	-0.0831***
		(0.0265)	(0.0265)
Threat × Landfalls Since 1980			0.0662*
			(0.0319)
Landfall × Landfalls Since 1980			-0.0653***
			(0.0121)
Fit statistics			
Observations	448,123	448,123	448,123
R^2	0.77057	0.77291	0.77291

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

are not significantly different from one another.

5.2 Regression Results

While the event studies above fail to show any significant impact of both additional hurricane exposure and additional timing between hurricanes for broad categories, it is still possible that the incremental changes are significant. I continue my analysis by estimated fixed-effects regression model in equation 2 for both additional hurricane landfalls, and additional years between subsequent hurricanes.

I start by estimating the effect of additional hurricane landfalls on sales. Table 1 shows the results from this specification. Column (1) shows the results of the base regression which excludes the historic county level variables. This column represents the average response of all counties. Although there is not a significant response to a hurricane threat, sales of bottled water increase by roughly 56% when there is a hurricane landfall.

Table 2: Recent Exposure Results

Dependent Variable:	log(total_rev_per_cap)		
Model:	(1)	(2)	(3)
Variables			
Hur_Threat	0.1073	-0.0129	1.155***
	(0.1564)	(0.1616)	(0.1954)
Hur_Landfall	0.4481***	0.5639***	-0.5641**
	(0.0683)	(0.0653)	(0.1926)
temp_mean	0.0046***	0.0066***	0.0066***
	(0.0010)	(0.0010)	(0.0010)
total_hist_landfall		-0.0779**	-0.0779**
		(0.0290)	(0.0290)
years_since_landfall_hur		0.0065	0.0065
		(0.0078)	(0.0078)
Hur_Threat × years_since_landfall_hur			-0.1341***
			(0.0220)
Hur_Landfall × years_since_landfall_hur			0.1278***
			(0.0192)
Fixed-effects			
year	Yes	Yes	Yes
month	Yes	Yes	Yes
fips	Yes	Yes	Yes
Fit statistics			
Observations	448,123	217,176	217,176
R^2	0.77057	0.79157	0.79158

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Column (2) shows the results from adding total hurricane landfalls since 1980 to the regression. It is clear that total exposure to hurricanes significantly affect not only the scale of purchases, but also the timing of such purchases. As a county experience additional hurricanes, sales shift to earlier in the forecast process. Specifically, there is a 6% decrease in sales associated with landfall, that are balanced by an equal 6% increase in sales from threats. While the event study results in figure 6 were week, they do visually follow this pattern of an increase in sales before a hurricane and a decrease in sales during the hurricane week.

Next, I analyze the effect of additional years between subsequent hurricanes. Table **??** shows the results from both equations 2 and 3. In addition to controlling for temperature in these regressions, I also control for the total historical landfalls from above. The result is a significant decrease in early sales by roughly 13% with an equal increase in sales closer to landfall. This is consistent with the results observed in the event study, and with the notion of fading memory.

5.3 Robustness

I test for the quadratic relationship of years between subsequent hurricanes and sales using the model in equation 3. Table **??** column (3) shows the results from this specification. Adding the squared term affects the baseline holdings of water, but does not affect the sales during a hurricane forecast or landfall.

6 Conclusion

As climate change continues, the intensity of natural disasters and extent of their destruction are expected to increase. Because this destruction also inhibits relief efforts, it is crucial that people know how to prepare for such disasters so that this delay is not harmful. I find that as counties are exposed to more hurricanes, the sales of emergency supplies actually increase in the presence of a storm. This indicates that people in these counties are learning how to prepare for disasters and adjusting. Additionally, the sales of emergency supplies decline in a way that follows the pattern of risk salience that is seen in other parts of the natural disaster literature when people are faced with a recurring disaster. The combination of needing experience and recent storms to purchase larger quantities of emergency supplies could leave areas with less experience or without a recent storm unprepared when faced with a hurricane. To help reduce the strain on relief efforts during future disasters, it be useful to find ways to bridge these gaps in emergency purchases.

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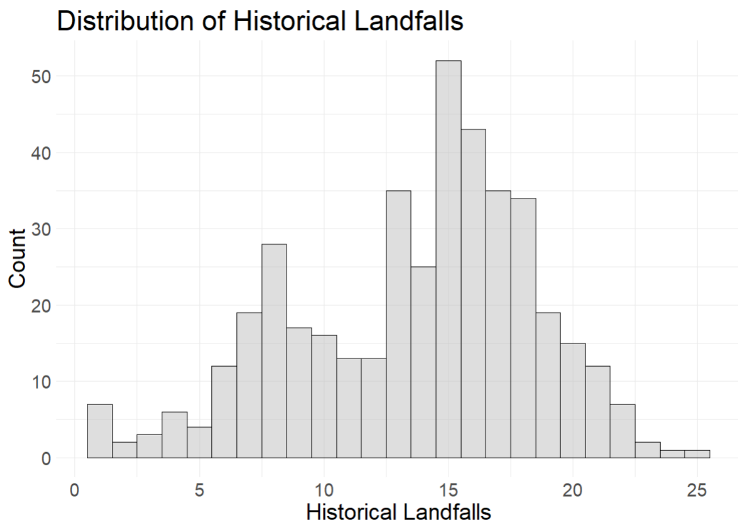
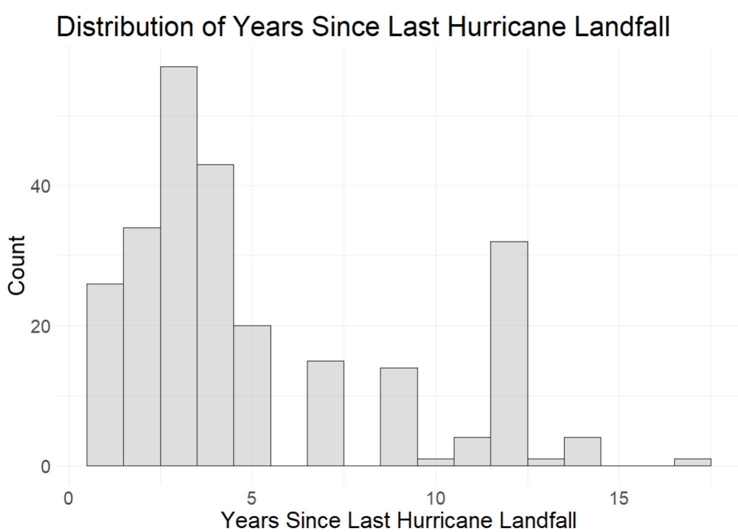
Appendix

Figure A1: Histogram of Observation Level Total Landfalls



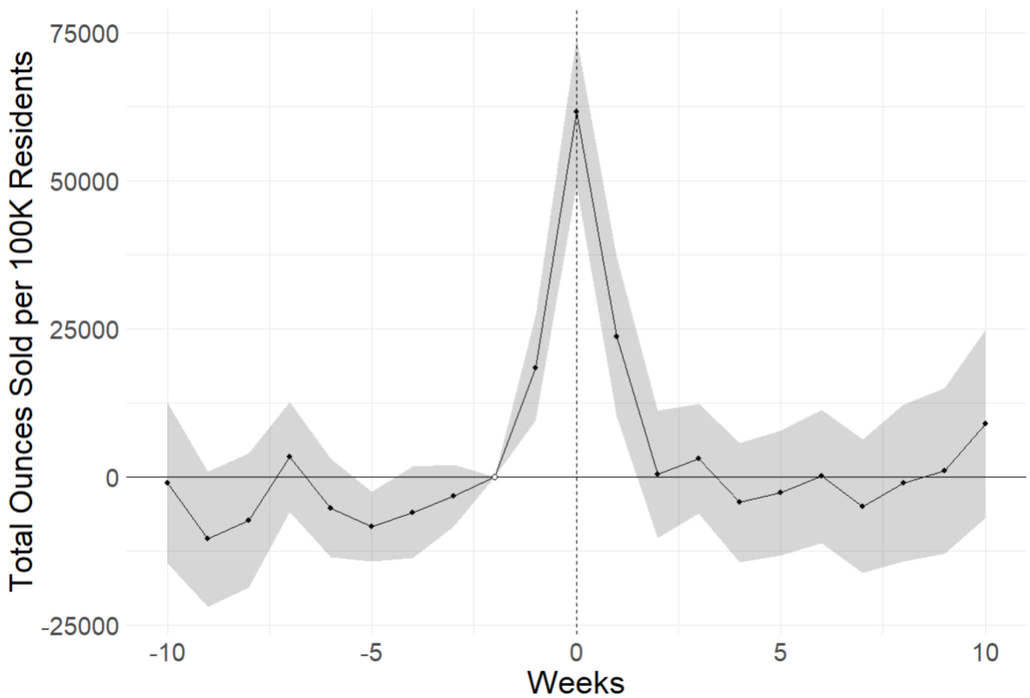
Note:

Figure A2: Histogram of Years Betwen Hurricanes



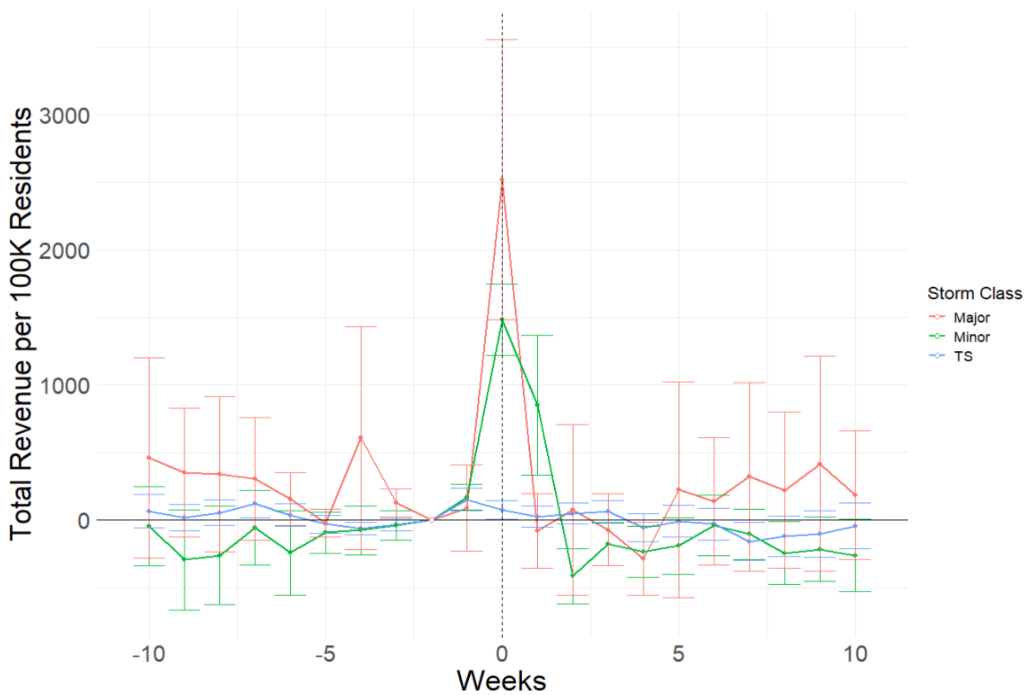
Note:

Figure A3: Event Studies by Hurricane Category



Note: Results from event study with subsets based on hurricane category. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars around each point represents the 95% confidence interval.

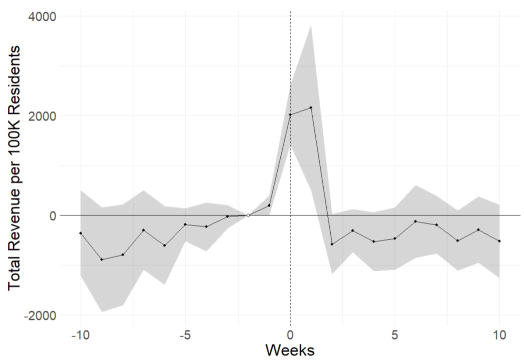
Figure A4: Event Studies by Hurricane Category



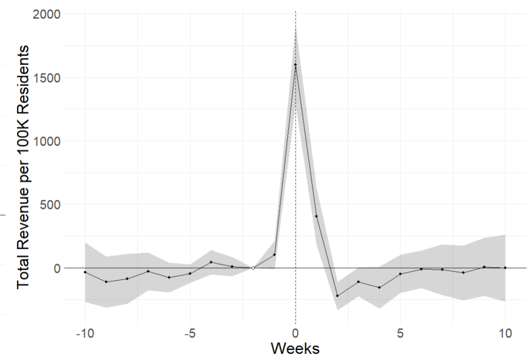
Note: Results from event study with subsets based on hurricane category. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars around each point represents the 95% confidence interval.

Figure A5: Event Studies By Historical Landfalls

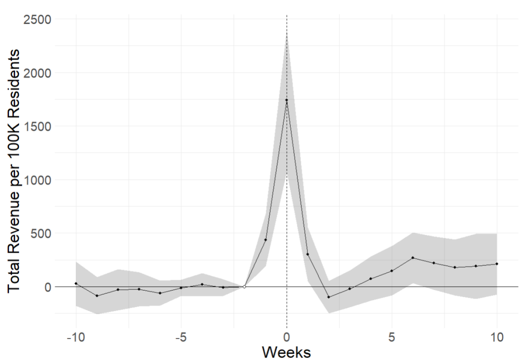
(a) 1 to 10 Landfalls



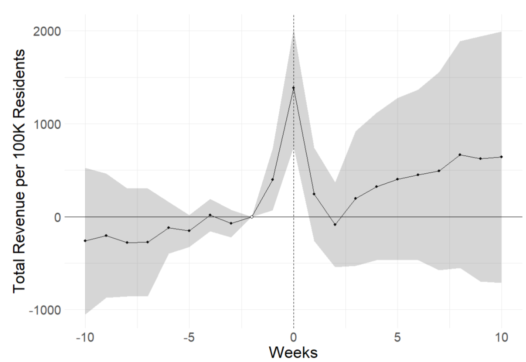
(b) 11 to 15 Landfalls



(c) 16 to 17 Landfalls

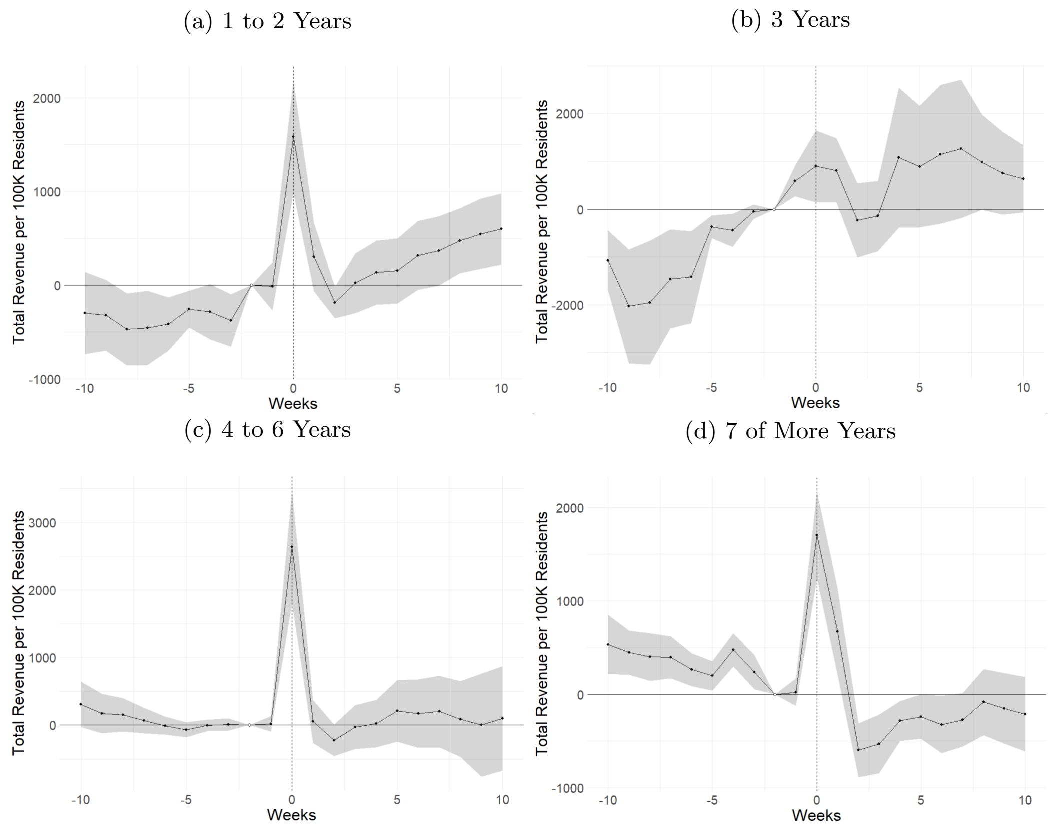


(d) 18 to 25 Landfalls



Note: Results from event study with subsets based on the total count of prior hurricanes to hit the county since 1980. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars represent the 5% confidence interval around the point estimate.

Figure A6: Event Studies By Years Between Subsequent Hurricanes



Note: Results from event study with subsets based on the years that have passed since that last storm made landfall in a county. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars represent the 5% confidence interval around the point estimate.

Table A1: Base Results

Dependent Variable:	Log Sales	
Model: All Storms	Hurricanes	
Variables		
Threat	0.0401*	0.1073
	(0.0193)	(0.1564)
Landfall	0.1962***	0.4481***
	(0.0606)	(0.0683)
temp_mean	0.0046***	0.0046***
	(0.0010)	(0.0010)
Fit statistics		
Observations	448,123	448,123
R^2	0.77052	0.77057

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

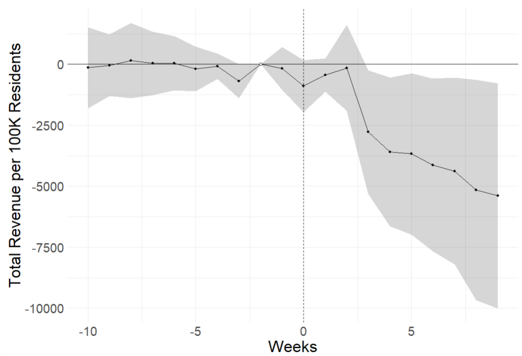
Table A2: Recent Exposure Results

Dependent Variable:	log(total_rev_per_cap)					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Hur_Threat	0.1073	-0.0129	1.155***	1.271***	-0.0162	1.435**
	(0.1564)	(0.1616)	(0.1954)	(0.1938)	(0.2339)	(0.5516)
Hur_Landfall	0.4481***	0.5639***	-0.5641**	-0.7123***	0.5344***	-0.9166
	(0.0683)	(0.0653)	(0.1926)	(0.1918)	(0.0678)	(0.5616)
temp_mean	0.0046***	0.0066***	0.0066***	0.0066***	0.0066***	0.0066***
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)
total_hist_landfall		-0.0779**	-0.0779**	-0.0893**	-0.0893**	-0.0893**
		(0.0290)	(0.0290)	(0.0303)	(0.0303)	(0.0303)
years_since_landfall_hur		0.0065	0.0065	0.0418**	0.0418**	0.0418**
		(0.0078)	(0.0078)	(0.0143)	(0.0143)	(0.0143)
Hur_Threat × years_since_landfall_hur			-0.1341***	-0.1448***	0.0144	-0.1830
			(0.0220)	(0.0208)	(0.0613)	(0.1503)
Hur_Landfall × years_since_landfall_hur			0.1278***	0.1412***		0.1974
			(0.0192)	(0.0188)		(0.1466)
yrs_since_hur_sq				-0.0028**	-0.0028**	-0.0028**
				(0.0009)	(0.0009)	(0.0009)
Hur_Threat × yrs_since_hur_sq					-0.0013	0.0021
					(0.0040)	(0.0089)
Hur_Landfall × yrs_since_hur_sq						-0.0033
						(0.0085)
Fixed-effects						
year	Yes	Yes	Yes	Yes	Yes	Yes
month	Yes	Yes	Yes	Yes	Yes	Yes
fips	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	448,123	217,176	217,176	217,176	217,176	217,176
R^2	0.77057	0.79157	0.79158	0.79303	0.79301	0.79303

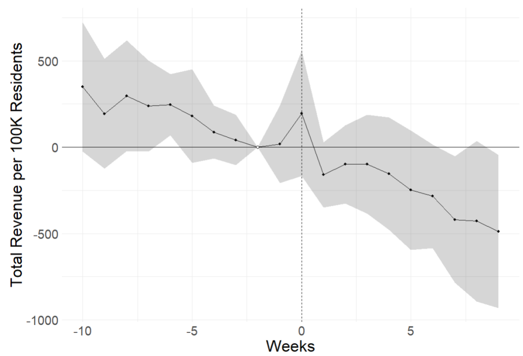
Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Figure A7: Placebo History

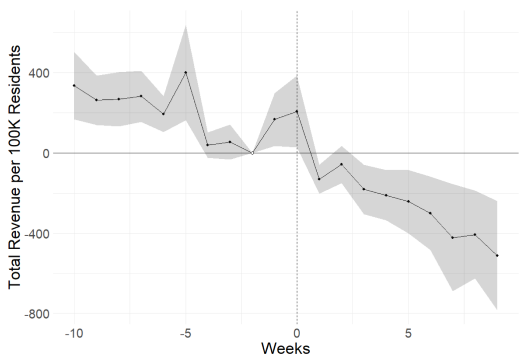
(a) 1 to 2 Years



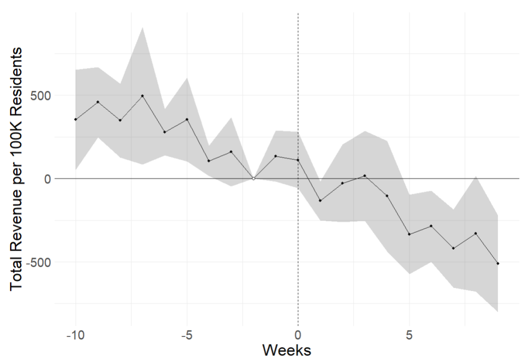
(b) 3 Years



(c) 4 to 6 Years

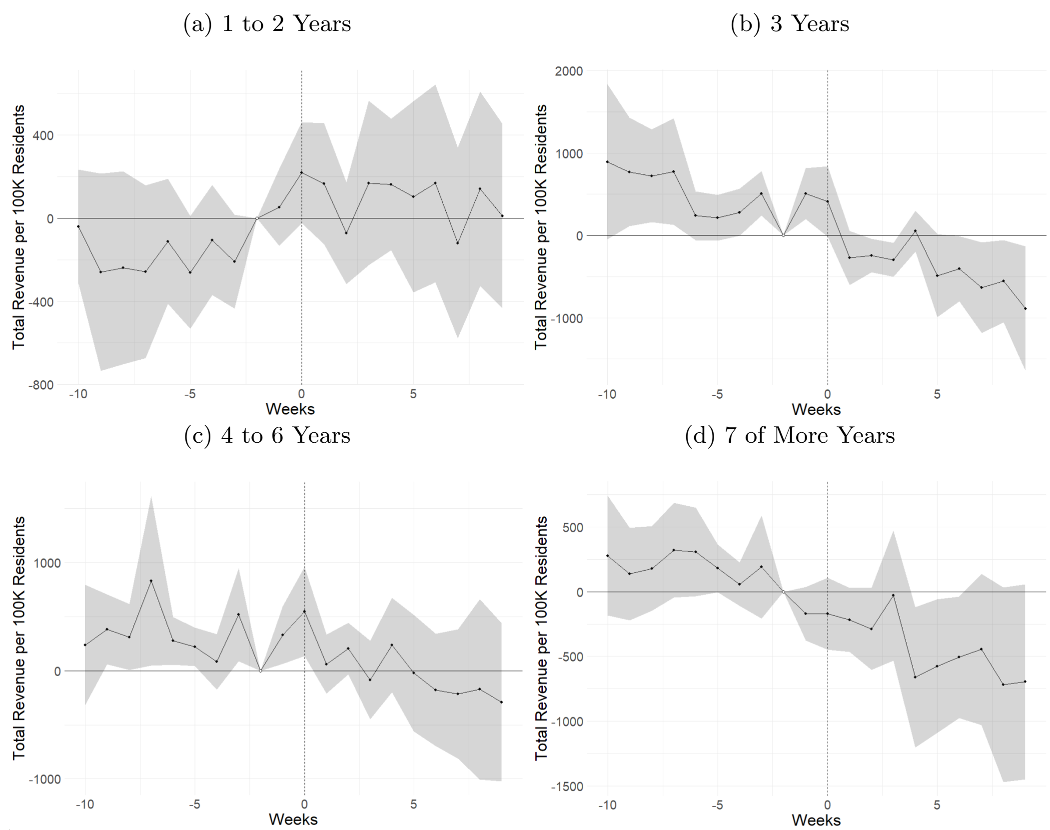


(d) 7 of More Years



Note: Results from event study with subsets based on the years that have passed since that last storm made landfall in a county. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars represent the 5% confidence interval around the point estimate.

Figure A8: Placebo By Years Between Subsequent Hurricanes



Note: Results from event study with subsets based on the years that have passed since that last storm made landfall in a county. The vertical axis represents total revenue per 100 thousand residents in USD. The horizontal axis represents weeks. Week 0 is the week that a tropical hurricane made landfall in a county. The bars represent the 5% confidence interval around the point estimate.

Table A3: Historical Exposure Robustness Test

Dependent Variable:	log(total_rev_per_cap)	
Model:	(1)	(2)
Variables		
plac_threat	0.0308	-0.2274
	(0.0642)	(0.2544)
plac_land	0.0748	0.2803*
	(0.0575)	(0.1311)
temp_mean	0.0057***	0.0057***
	(0.0007)	(0.0007)
total_hist_landfall	-0.0767*	-0.0768*
	(0.0358)	(0.0358)
plac_threat × total_hist_landfall		0.0181
		(0.0155)
plac_land × total_hist_landfall		-0.0142
		(0.0110)
Fixed-effects		
year	Yes	Yes
month	Yes	Yes
fips	Yes	Yes
Fit statistics		
Observations	407,459	407,459
R^2	0.79756	0.79756

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A4: Years Between Robustness Test

Dependent Variable:	log(total_rev_per_cap)	
Model:	(1)	(2)
Variables		
plac_threat	0.1025**	0.2976
	(0.0428)	(0.2509)
plac_land	0.0220	0.0297
	(0.0380)	(0.1570)
temp_mean	0.0073***	0.0073***
	(0.0007)	(0.0007)
years_since_landfall_hur	0.0121	0.0122
	(0.0083)	(0.0082)
plac_threat × years_since_landfall_hur		-0.0314
		(0.0407)
plac_land × years_since_landfall_hur		-0.0011
		(0.0177)
Fixed-effects		
year	Yes	Yes
month	Yes	Yes
fips	Yes	Yes
Fit statistics		
Observations	197,654	197,654
R^2	0.82215	0.82217

Clustered (fips & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*